Q1: (a)

i) - physical address of word = 0xB894A

- offset from VGA (B8000h) = 0x094A

```
  AF000
+  994A
 B894A
- B8000
 094A
```

- coordinate interm of (row, col) :

0x094A → decimal → $9\times16^2 + 4\times16^1 + 10\times16^0$

$= 2378$

$\frac{2376}{160}$; (to get row)

$= 14\ \frac{138}{160}$ ∴ Row = 14 (first row is 0)

$\frac{138}{2} = 69$ ∴ col = 69 (first col is 0)

⟹ Text displayed at (14, 69).

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ii)

ES = 0xB800

for DI : ((80 x row) + column) x 2

= ((80 x 14) + 35) x 2

= 2310

$(2310)_{10} \rightarrow (0906)_{16}$

DI = 0x0906

AX = 0x9E56

AH = 0x9E

AL = 0x56

attribute in hexa = 9Eh

" " binary = 1001 1110b

(b)

i) Process Data:

```
push bp,4
mov bp,sp
~~sub~~ ~~sp~~
sub sp, 4
push ax
~~push~~ bx
push si
/ // /
pop si
pop bx
pop ax
mov sp,bp
pop bp
ret 6
```

prologue

epilogue



(ii)

Initially SP = FFFEh

- After,

  3 parameters = FFFEh − 6h = FFF8h

  sub - called (ret addr) = FFF8h − 2h = FFF6h

  2 local - vars = FFF6h − 4h = FFF2h

  4 registers = FFF2h − 8h = FFEAh

⇒ After prologue SP = FFEAh

- After Epilogue,

  3 registers pop = FFEAh + 6h = FFF0h

  mov sp, bp = FFF0h + 4h = FFF4h

  pop bp = FFF4h + 2h = FFF6h

  ret 6 = FFF6h + 2h + 6h = FFFEh

⇒ After Epilogue SP = FFFEh

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(c)

i)

SI = 0102h + 08h

DI = 010Eh + 08h

CX = 0008h − 08h

SI +1 +1 +1 +1 +1 +1 ✱✱
DF +1
F A S T - N U - L H R O
DI +1 +1 +1 +1 +1 +1 ✱✱
F A S T - N U C E S O
CX ✱1 -1 -1 + + + -1 Break
-1

∴ SI = 010Ah

DI = 0116h

CX = 0003h

ii)

ZF = 0 ; as repe cmpsb will continue until zf is zero

CF = 1 ; by calculation

16 (borrowed)
2D
− 43
EA

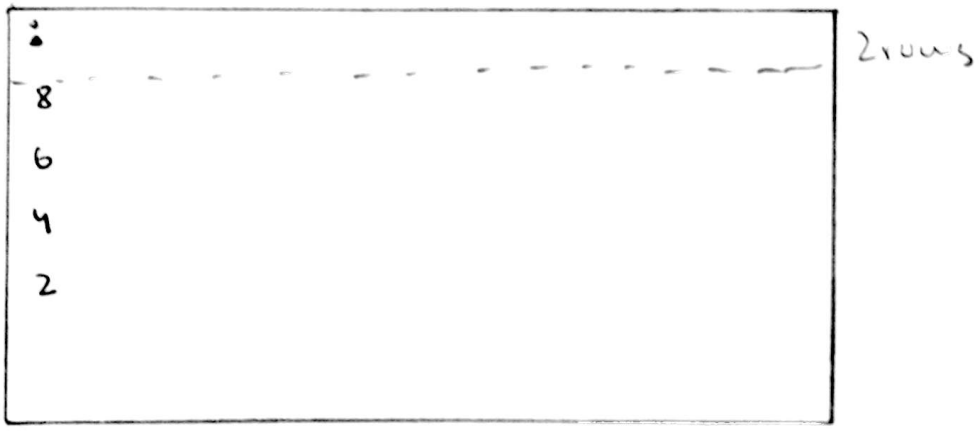
iii) CLD sets DF to 0 and STD sets to 1, CLD causes forword movement with SI & DI increments while STD results in the opposite. Thus both SI & DI will move backwords CX times until both have different values, which is more likely as they are random.

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(d)

Printing	CX	Char	DI	Row	Col
1	5	:	0	0	0
2	4	8	320	2	0
3	3	6	640	4	0
4	2	4	960	6	0
5	1	2	1280	8	0

* characters are light red on a black bg



(e) i)

```
ret
push dx
pop  ax
ret 4
```

ii)

```
[org 100h]
mov ax, 5
mov bx, 3
push ax
push bx
pop  ax
pop  bx
mov  ax, 4c00h
int  21h
```

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